Lesson 2: Reference Frames

MECH 637: Astrodynamic Reentry
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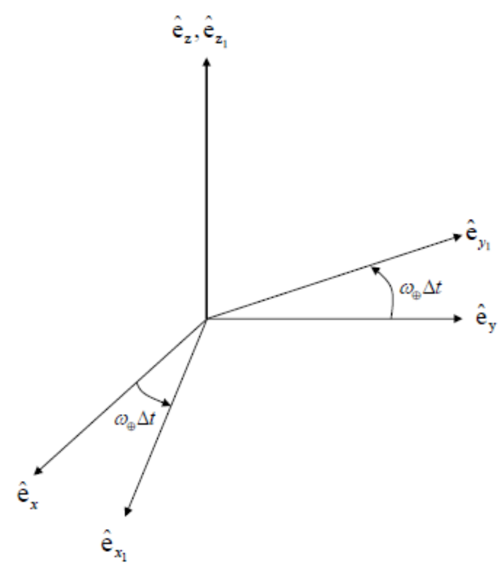
Geocentric-Equatorial Coordinate Frame

- Frame notation: OXYZ
- Inertial reference frame with origin at planetary center
- x-axis $\left(\hat{e}_x\right)$: Points in direction of the Vernal Equinox
- z-axis $\left(\hat{e}_z\right)$: Points in direction of the North Pole
- y-axis $\left(\hat{e}_y\right)$: Orthogonal to the x-, z-axes; lies in the equatorial plane: $\hat{e}_z = \hat{e}_x \times \hat{e}_y$

Planet-Fixed Coordinate Frame



- Frame notation: $\mathrm{OX_1Y_1Z_1}$
- Planet-fixed reference frame with origin at planetary center
- z_1-axis $\left(\hat{e}_{z_1}\right)$: Points in direction of the North Pole
- x_1, y_1 plane coincident with equatorial plane
- Assume planet rotates with a constant velocity: $\vec{\omega}_\oplus = \omega_\oplus \hat{e}_{z_1} = \omega_\oplus \hat{e}_z$
- Coordinate transformation, where $\hat{\mathbf{e}}$ and $\hat{\mathbf{e}}_1$ represent the unit vectors of the OXYZ and $\mathrm{OX_1Y_1Z_1}$ frames, respectively:

$$\left[\hat{\mathbf{e}}_1\right] = \begin{bmatrix} \cos\left(\omega_{\oplus}\,\Delta\,t\right) & \sin\left(\omega_{\oplus}\,\Delta\,t\right) & 0 \\ -\sin\left(\omega_{\oplus}\,\Delta\,t\right) & \cos\left(\omega_{\oplus}\,\Delta\,t\right) & 0 \\ 0 & 0 & 1 \end{bmatrix} \left[\hat{\mathbf{e}}\right] = \mathbf{R}_{\mathbf{z}}\left(\omega_{\oplus}\,\Delta\,t\right)\left[\hat{\mathbf{e}}\right]$$

Where $\mathbf{R}_{\mathbf{z}}$ represents the rotation matrix about the z-axis.

Vehicle-Pointing Coordinate Frame

- Frame notation: $\mathrm{OX_2Y_2Z_2}$
- Origin at planetary center
- x_2-axis $\left(\hat{e}_{x_2}\right)$: Points along vehicle's position vector
- y_2-axis $\left(\hat{e}_{y_2}\right)$: Parallel to equatorial plane
- z_2-axis $\left(\hat{e}_{z_2}\right)$: Orthogonal to the x_2-, y_2-axes and completes right-handed system
- Coordinate transformation, where $\hat{\mathbf{e}}_1$ and $\hat{\mathbf{e}}_2$ represent the unit vectors of the $\mathrm{OX_1Y_1Z_1}$ and $\mathrm{OX_2Y_2Z_2}$ frames, respectively:

$$\left[\hat{\mathbf{e}}_2\right] = \begin{bmatrix} \cos(-\phi) & 0 & -\sin(-\phi) \\ 0 & 1 & 0 \\ \sin(-\phi) & 0 & \cos(-\phi) \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \left[\hat{\mathbf{e}}_1\right]$$

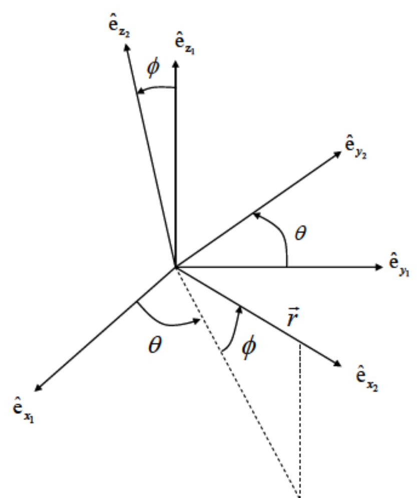
$$= \begin{bmatrix} \cos\phi\cos\theta & \cos\phi\sin\theta & \sin\phi \\ -\sin\theta & \cos\theta & 0 \\ -\sin\phi\cos\theta & -\sin\phi\sin\theta & \cos\phi \end{bmatrix} \left[\hat{\mathbf{e}}_1\right] = \mathbf{R}_{\mathbf{y}_2}(-\phi)\mathbf{R}_{\mathbf{z}_1}(\theta)\left[\hat{\mathbf{e}}_1\right]$$

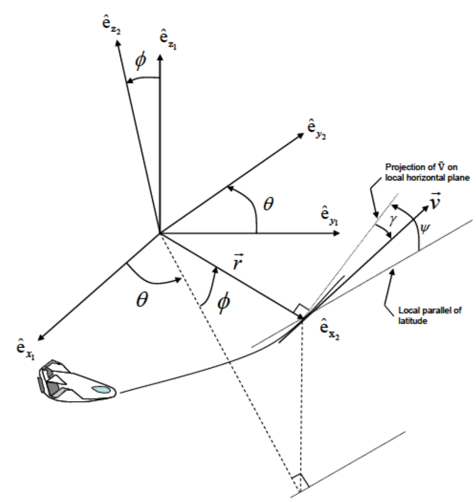
Where θ is longitude and ϕ is latitude. *Note: Coordinate transformation is accomplished by first multiplying $\hat{\boldsymbol{e}}_1$ by $\boldsymbol{R}_{z_1}(\theta)$, then the result by $\boldsymbol{R}_{y_2}(-\phi)$.*

- Coordinate transformation, where $\hat{\mathbf{e}}$ and $\hat{\mathbf{e}}_2$ represent the unit vectors of the OXYZ and $\mathrm{OX_2Y_2Z_2}$ frames, respectively:

$$\left[\hat{\mathbf{e}}_2\right] = \mathbf{R}_{\mathbf{y}_2}(-\phi)\mathbf{R}_{\mathbf{z}_1}(\theta)\mathbf{R}_{\mathbf{z}}\left(\omega_{\oplus}\,\Delta\,t\right)\left[\hat{\mathbf{e}}\right]$$

- **Flight-Path Angle (γ):** The angle between the local horizontal frame (i.e., the plane passing through the vehicle and orthogonal to $\vec{r}$) and the velocity $\vec{v}$
    - γ is positive when $\vec{v}$ is above the local horizontal plane
    - γ describes how much the velocity vector contributes to moving "in and out" along the radial vector
- **Heading Angle (ψ):** The angle between the local parallel of latitude and the projection of $\vec{v}$ onto the local horizontal plane
    - ψ is positive in the right-handed direction about the x_2-axis
    - ψ describes how much the velocity vector contributes to moving "toward and away" from the equatorial plane



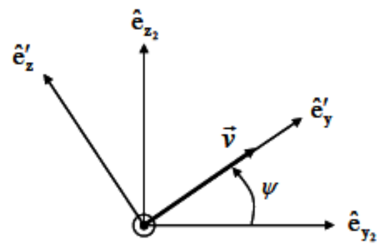


Velocity-Referenced Coordinate Frame

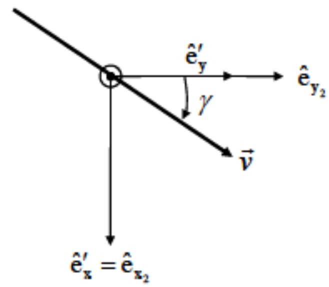
- Frame notation: $\mathrm{OX''Y''Z''}$
- *Step 1:* Create $\mathrm{OX'Y'Z'}$ by rotating the $\mathrm{OX_2Y_2Z_2}$ frame about the x_2-axis by ψ so that the "new" y-axis, y', is aligned with the projection of $\vec{v}$ on the local horizontal plane:

$$\begin{bmatrix} \hat{e}_x' \\ \hat{e}_y' \\ \hat{e}_z' \end{bmatrix} = \begin{bmatrix} \hat{e}_{x2} \\ \hat{e}_y' \\ \hat{e}_z' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\psi & \sin\psi \\ 0 & -\sin\psi & \cos\psi \end{bmatrix} \begin{bmatrix} \hat{e}_{x2} \\ \hat{e}_{y2} \\ \hat{e}_{z2} \end{bmatrix}$$

$$\left[\hat{\mathbf{e}}'\right] = \mathbf{R}_{\mathbf{x}_2}(\psi)\left[\hat{\mathbf{e}}_2\right]$$

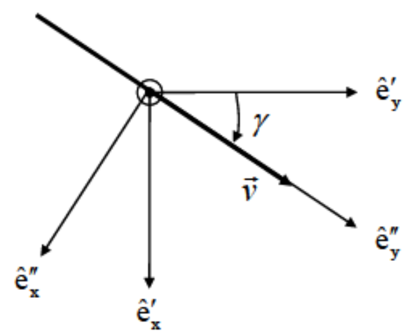


Rotation by ψ as seen looking down the x_2-axis



Rotation by ψ as seen looking down the z_2-axis (ψ obscured by $\hat{e}_{y_2}$ axis)

- *Step 2:* ***Negatively*** rotate the $OX'Y'Z'$ frame about the z'-axis by γ so that the "new" y-axis, y'', is aligned with $\vec{v}$:



Negative rotation by γ as seen looking down the z′-axis

$$\begin{bmatrix} \hat{e}_x'' \\ \hat{e}_y'' \\ \hat{e}_z'' \end{bmatrix} = \begin{bmatrix} \hat{e}_x'' \\ \hat{e}_y'' \\ \hat{e}_z' \end{bmatrix} = \begin{bmatrix} \cos(-\gamma) & \sin(-\gamma) & 0 \\ -\sin(-\gamma) & \cos(-\gamma) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{e}_x' \\ \hat{e}_y' \\ \hat{e}_z' \end{bmatrix}$$

$$\begin{bmatrix} \hat{e}_x'' \\ \hat{e}_y'' \\ \hat{e}_z'' \end{bmatrix} = \begin{bmatrix} \hat{e}_x'' \\ \hat{e}_y'' \\ \hat{e}_z' \end{bmatrix} = \begin{bmatrix} \cos\gamma & -\sin\gamma & 0 \\ \sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{e}_x' \\ \hat{e}_y' \\ \hat{e}_z' \end{bmatrix}$$

$$\left[\hat{\mathbf{e}}''\right] = \mathbf{R}_{z'}(-\gamma)\left[\hat{\mathbf{e}}'\right]$$

- Final coordinate transformation:

$$\begin{bmatrix} \hat{e}_x'' \\ \hat{e}_y'' \\ \hat{e}_z'' \end{bmatrix} = \begin{bmatrix} \cos\gamma & -\sin\gamma\cos\psi & -\sin\gamma\sin\psi \\ \sin\gamma & \cos\gamma\cos\psi & \cos\gamma\sin\psi \\ 0 & -\sin\psi & \cos\psi \end{bmatrix} \begin{bmatrix} \hat{e}_{x2} \\ \hat{e}_{y2} \\ \hat{e}_{z2} \end{bmatrix}$$

$$\left[\hat{\mathbf{e}}''\right] = \mathbf{R}_{z'}(-\gamma)\mathbf{R}_{\mathbf{x}_2}(\psi)\left[\hat{\mathbf{e}}_2\right]$$

  - $\hat{e}_y''$-axis: Points along the velocity vector $\vec{v}$; equivalently denoted as $\hat{e}_v$

- $\hat{e}_z''$-axis: Perpendicular to both the velocity $\vec{v}$ and radial $\vec{r}$ vectors, and lies in the local horizontal plane

Relative Angular Motion

- Angular rotation is terms of the $\mathrm{OX_1Y_1Z_1}$ and OXYZ frames, respectively:

$$\vec{\omega}_{\oplus} = \omega_{\oplus}\hat{e}_{z_1} = \omega_{\oplus}\hat{e}_z$$

Recall the transformation from the $\mathrm{OX_1Y_1Z_1}$ to the $\mathrm{OX_2Y_2Z_2}$ frame:

$$\left[\hat{\mathbf{e}}_2\right] = \begin{bmatrix} \cos\phi\cos\theta & \cos\phi\sin\theta & \sin\phi \\ -\sin\theta & \cos\theta & 0 \\ -\sin\phi\cos\theta & -\sin\phi\sin\theta & \cos\phi \end{bmatrix} \left[\hat{\mathbf{e}}_1\right]$$

$$\left[\vec{\boldsymbol{\omega}}\right]_{1/0}^{\hat{\mathbf{e}}_2} = \begin{bmatrix} \cos\phi\cos\theta & \cos\phi\sin\theta & \sin\phi \\ -\sin\theta & \cos\theta & 0 \\ -\sin\phi\cos\theta & -\sin\phi\sin\theta & \cos\phi \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \omega_{\oplus} \end{bmatrix}$$

$$\vec{\omega}_{\oplus} = \left[\vec{\boldsymbol{\omega}}\right]_{1/0}^{\hat{\mathbf{e}}_2} = \left(\omega_{\oplus}\sin\phi\right)\hat{e}_{x_2} + \left(\omega_{\oplus}\cos\phi\right)\hat{e}_{z_2}$$

Note: $\left[\vec{\boldsymbol{\omega}}\right]_{1/0}^{\hat{\mathbf{e}}_2}$ *indicates that the angular rate of* $\mathrm{OX_1Y_1Z_1}$ *relative to* OXYZ *is written in terms of coordinates along the* $\left[\hat{\mathbf{e}}_2\right]$ *unit vectors.*

- An equation for the overall angular rate can be assembled according to the transformation to $\left[\hat{\mathbf{e}}_2\right]$ from $\left[\hat{\mathbf{e}}_1\right]$ unit vectors:

$$\vec{\Omega} = \dot{\theta}\hat{e}_{z_1} - \dot{\phi}\hat{e}_{y_2}$$

When expressed entirely in the vehicle-pointing frame, the preceding equation becomes:

$$\left[\vec{\boldsymbol{\Omega}}\right]_{2/1}^{\hat{\mathbf{e}}_2} = \vec{\Omega} = \left(\dot{\theta}\sin\phi\right)\hat{e}_{x_2} - \dot{\phi}\hat{e}_{y_2} + \left(\dot{\theta}\cos\phi\right)\hat{e}_{z_2}$$

Source Consulted

Hicks, Kerry D. *Introduction to Astrodynamic Reentry*, Second Edition. Dayton, OH: Hicks, 2014.